How-To Library

Informatica

How to configure an Azure hosted PostgreSQL database with the Process Server

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Abstract

This article describes how you can connect an Azure hosted PostgreSQL database to the Process Server with a custom user and custom database. It also describes how you can migrate the existing data from the Process Server local PostgreSQL database to the Azure hosted PostgreSQL database.

Supported Versions

- Azure hosted PostgreSQL database version 18.4
- PostgreSQL database version 18.4.2026.10.M

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Overview

Process Server is the Secure Agent service that executes Application Integration processes, connectors, and connections.

When you deploy Application Integration assets to the Secure Agent, you deploy them to the Process Server. When you run an asset, the Process Server executes it.

The PostgreSQL database that the Process Server runs on stores the metadata that the Process Server collects and generates.

Informatica includes a PostgreSQL database with the Process Server package. However, you can also connect a different PostgreSQL database version to the Process Server. You can also migrate the default Process Server PostgreSQL database dump to the Azure hosted PostgreSQL database.

Prerequisites

Verify that the following prerequisites are met before you connect an Azure hosted PostgreSQL database to the Process Server, and migrate the data from the default Process Server database to the Azure hosted PostgreSQL database:

- You must create a user. For example, user1.
- The user must have the following privileges:

  - rolsuper – role super-user

- rolinherit
- rolcreaterole
- rolcreatedb
- rolcanlogin
- rolreplication
- rolconnlimit
- rolbypassrls

The **Superuser** role is optional.

The following image shows the privileges that the user must have. The privileges might differ based on the PostgreSQL database:



- Do not use the Postgres utilities such as `db_reset` and `server_stop` that are shared with the Secure Agent package in the following directory:
  `<Secure Agent installation directory>\apps\process-engine\data\db\util`

  Use the related database utilities.
- Provision the sizing of a PostgreSQL instance on Azure based on your requirement with at least 100 GB of storage.
- Create a new database. For example, *user_db*. You can use the pgAdmin software to connect to the database, create users, and create databases.
- The character encoding of the database must be UTF-8 and the database server parameters must contain the following parameters:

```
statement_timeout = 0; lock_timeout = 0; client_encoding = 'UTF8';
standard_conforming_strings = on;
```

Connecting an Azure hosted PostgreSQL database to the Secure Agent

To connect an Azure hosted PostgreSQL database to the Secure Agent installed on your machine, perform the following steps:

Step 1: Install the CAI postgres related database, schemas, and privileges

Step 2: Connect the Process Server to the Azure hosted PostgreSQL database

Step 1. Install the CAI postgres related database, schemas, and privileges

The Process Server inserts data into specific tables such as the aeengine table and process engine table, and their respective columns. You must manually create similar tables and a user (<user_name>) in the Azure hosted PostgreSQL database.

To create tables, schemas, and privileges, perform the following steps:

1. Log in to the Secure Agent machine.
2. Go to the following directory:

   `<Secure Agent installation directory>\apps\process-engine\data\db\util\sql`
3. Open the `schema.sql` file in a text editor and comment out line numbers 26 and 33 as shown below:

   ```
   CREATE EXTENSION IF NOT EXISTS plpgsql WITH SCHEMA pg_catalog;
   COMMENT ON EXTENSION plpgsql IS 'PL/pgSQL procedural language';
   ```
4. In the `schema.sql` file, search for the following lines, and replace `bpeluser` with the <user_name> and replace `activevos` database name with the <user_db> name:

   ```
   DROP DATABASE IF EXISTS activevos;
   CREATE DATABASE activevos ENCODING utf8;
   \connect activevos
   REVOKE ALL ON SCHEMA public FROM bpeluser;
   GRANT ALL ON SCHEMA public TO bpeluser;
   ```

   In case you have created the database manually, you might want to create only the tables. In that case, you can comment out the following lines:

   ```
   DROP DATABASE IF EXISTS activevos;
   CREATE DATABASE activevos ENCODING utf8;
   ```

   Then update your <user_db> name and <user name> in the remaining lines.
5. Go to the following directory:

   `<Secure Agent installation directory>\apps\process-engine\data\db\postgresql-windows-x64-binaries\pgsql\bin`
6. On Windows, open the command prompt in the Administrator mode.
7. Run the database script using the following command:

   ```
   psql.exe -U <username> -p <port_number> -h <host_name> -f "<Secure Agent installation directory>\apps\process-engine\data\db\util\sql\schema.sql" -d postgres
   ```

   Here, `host_name` is the Azure hosted PostgreSQL database server name and `port_number` is the database port. The user name and password are the credentials of the new user.

   After the command completes, you see that the <user_db> database is created in the Azure hosted PostgreSQL database with all the tables and necessary privileges.

   After the database is created, you must create new environment variables.

To create new environment variables, perform the following steps:

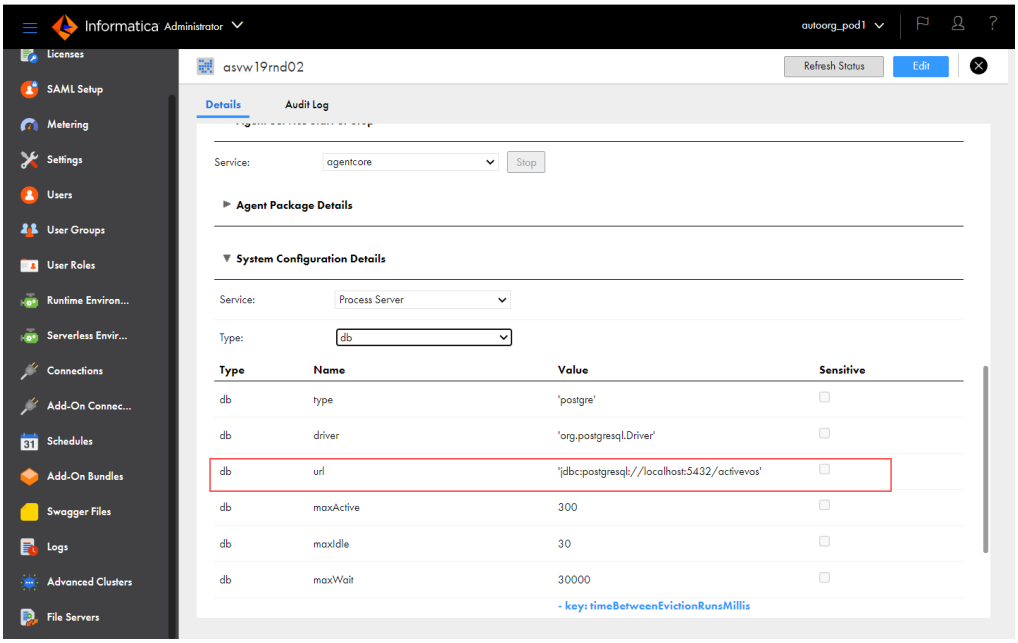
1. Set new environment variable `PE_DB_USERNAME` and enter the value as <user_name>.
2. Set new environment variable `PE_DB_PASSWORD` and enter the password for the <user_name>.
3. Save the variables.
4. Add the new environment variables to the path variable.

These steps are for the Windows environment. Perform similar steps for the Linux environment.

Step 2. Connect the Process Server to the Azure hosted PostgreSQL database

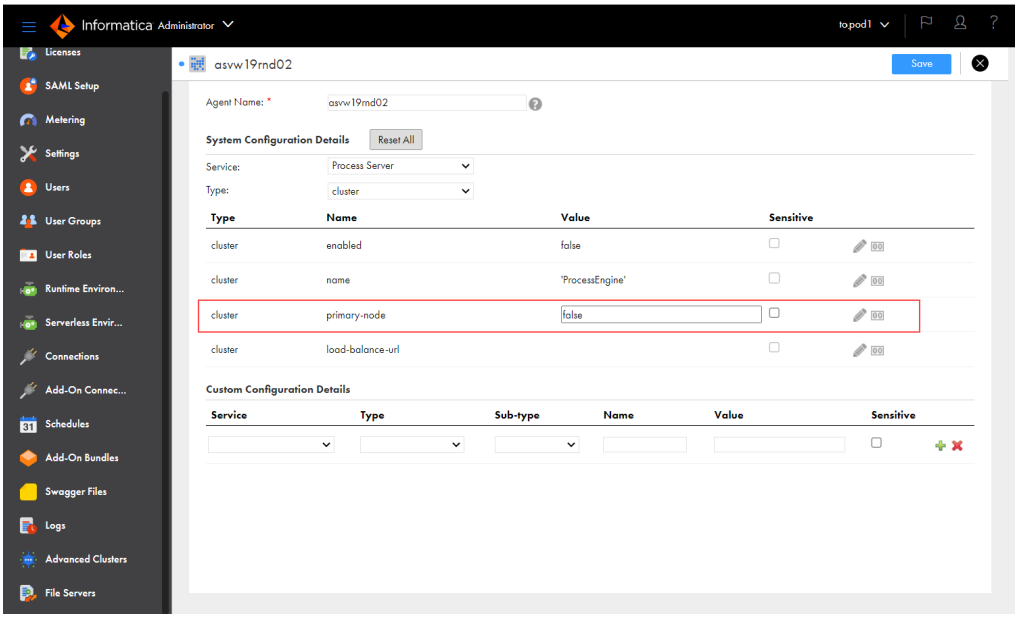
After the Process Server is normally up and running on your system, perform the following steps:

1. In the **Administrator** service, on the **Runtime Environments** page, click the name of the Secure Agent.
   You might have to expand the Secure Agent group to see the list of Secure Agents within the group.
2. Click the **Details** tab.
3. In the upper right corner, click **Edit**.
4. To edit a service property, perform the following steps:
   a. In the **System Configuration Details** area, select the service as **Process Server**.
   b. Select the configuration property type as **db**.
   c. In the row that contains the **Name** as **url**, click the **Edit Agent Configuration** icon and enter `hostname:port_number` as shown in the following image:



   Here, `hostname` is the Azure hosted PostgreSQL database server name and `port` is the database port. Ensure that you enter the `localhost` as the `hostname` of the Azure hosted PostgreSQL database server name, and replace `activevos` with your database name, for example, <`user_db`>.
   d. Select the configuration property type as **cluster**.

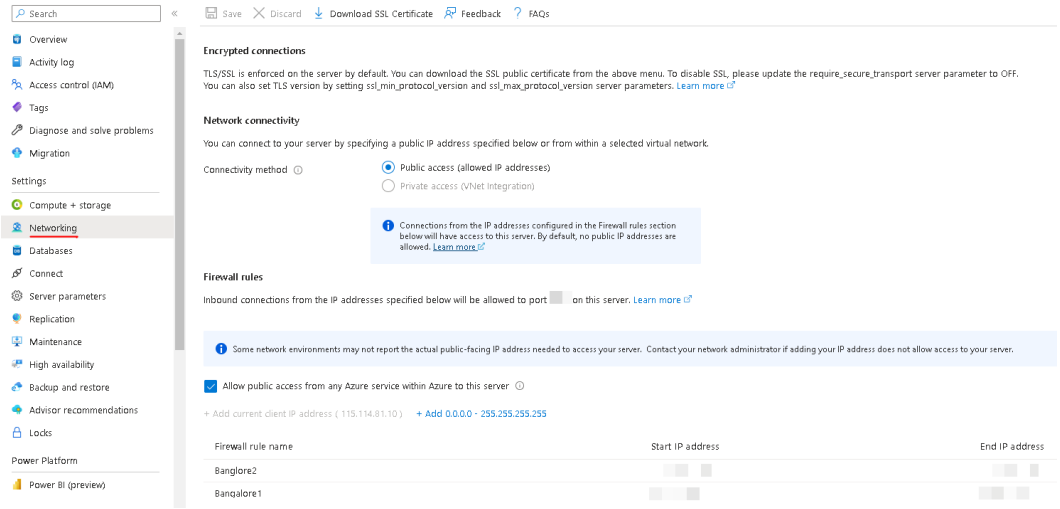
e. In the row that contains the **Name** as **primary-node**, click the **Edit Agent Configuration** icon, and update the value to **false** as shown in the following image:



f. Click **Save**.

The Process Server will be in the restart required state after making the changes to the system configuration.

5. Add your machine IP address where the Secure Agent is installed to the firewall rules of Networking in the database portal as shown in the following image:



6. Log in to the Secure Agent machine.
7. Go to the following directory:

   `<Secure Agent installation directory>\apps\agentcore`
8. Run `infaagent.sh shutdown`.
9. Go to the following directory:

   `<Secure Agent installation directory>\apps\process-engine\data\db\util`
10. Run `server_stop`.

11. To start the Secure Agent, go to the following directory:

    ```
    <Secure Agent installation directory>\apps\agentcore
    ```

12. Run `infaagent.sh startup`.

    To verify that your user name and password are being used, after the Process Server is up, verify that the `-Dpe.db.username` value is set to your user name in the Catalina logs available in the following directory:

    ```
    <Secure Agent installation directory>\apps\process-engine\logs
    ```

    It might take some time for the Process Server to restart depending upon the latency between the Secure Agent machine and the PostgreSQL database.

Migrating the existing cluster database dump to the Azure hosted PostgreSQL database

To migrate data from the existing cluster database dump to the Azure hosted PostgreSQL database, perform the following steps:

Step 1: Capture the database dump from the existing Process Server cluster database

Step 2: Migrate the database dump to the Azure hosted PostgreSQL database

Step 3: Connect the Process Server to the Azure hosted PostgreSQL database

Step 1. Capture the database dump from the existing Process Server cluster database

To capture the database dump from the existing Process Server cluster database, perform the following steps:

1. In the **Application Integration Console** service, on the **Process Schedules** page, disable the schedules on the Secure Agent.
2. Select all the enabled schedules, and from the drop-down list, click **Disable**.
3. In the **Administrator** service, on the **Runtime Environments** page, click the name of the Secure Agent.
4. Click the **Details** tab.

   You might have to expand the Secure Agent group to see the list of Secure Agents within the group.
5. In the **Agent Service Start or Stop** area, select the service as **Process Server** and click **Stop**.
6. Use a non-root user to take the Process Server PostgreSQL database backup.
7. To ensure that the Process Server cluster PostgreSQL database is up and running, run the following command from the machine where the Secure Agent is installed:

   ```
   ps -ef|grep postgres
   ```

8. Run the `db_backup` script from the following path:

   ```
   \home\Informatica Cloud Secure Agent\apps\process-engine\data\db\util > ./db_backup.sh
   <username> <password> <fileName>.dump
   ```

   The user name is the default user name *bpeluser* and the password is *bpel*.
9. Enter the default user name and password of the Process Server PostgreSQL database.

Step 2. Migrate the database dump to the Azure hosted PostgreSQL database

You can use the `pg_restore` script to restore the captured database dump to an Azure hosted PostgreSQL database.

To migrate the DB dump to the Azure hosted PostgreSQL database, perform the following steps:

1. Log in to the Secure Agent machine.
2. Go to the following directory:

   `<Secure Agent installation directory>/apps/process-engine/data/db/postgresql-linux-x64-binaries/pgsql/bin`
3. On Linux, open the command prompt.
4. Run the `pg_restore` script using the following command:

   ```
   ./pg_restore -d <dbname> -h <hostname> -U <dbuser> <db dump path with file name.dump> -p 5432
   ```

   Here, `dbname`, `hostname`, and `dbuser` (non-root user) are the database name, host name, and database user details of the Azure hosted PostgreSQL database. The `dbname` is your database name.
5. Enter the password when prompted.

   The password is the user password.

   The database is successfully migrated.

   You might encounter the following error:

   ```
   pg_restore: error: connection to database "activevos" failed: SCRAM
   authentication requires libpq version10 or above
   ```

   To resolve this issue, perform the following steps:

   1. Go to the following directory:

      `<Secure Agent installation directory>/apps/process-engine/data/db/postgresql-linux-x64-binaries/pgsql/lib`
   2. Run the following command:

      ```
      export LD_LIBRARY_PATH=`pwd`
      ```
   3. Go to the following directory:

      `<Secure Agent installation directory>/apps/process-engine/data/db/postgresql-linux-x64-binaries/pgsql/bin`
   4. Run the `pg_restore` script using the following command:

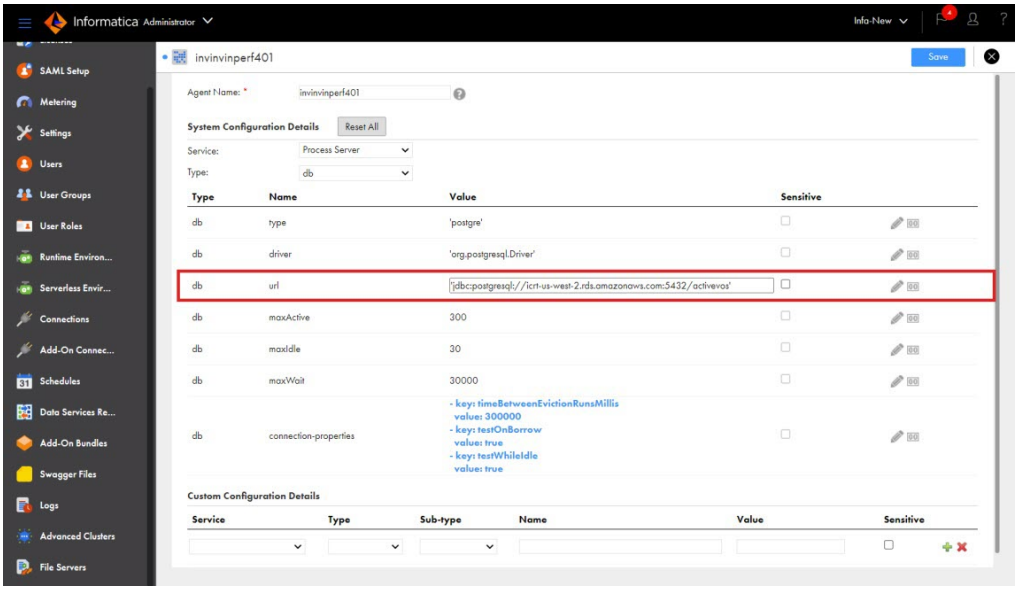
      ```
      ./pg_restore -d <dbname> -h <hostname> -U <dbuser> <db dump path with file name.dump> -p 5432
      ```
6. In the **Administrator** service, on the **Runtime Environments** page, select the Secure Agent.
7. In the **Agent Service Start or Stop** area, select the service as **Process Server**.
8. Click **Start** to start the Process Server.

Step 3. Connect the Process Server to the Azure hosted PostgreSQL database

After the Process Server is normally up and running on your system, perform the following steps:

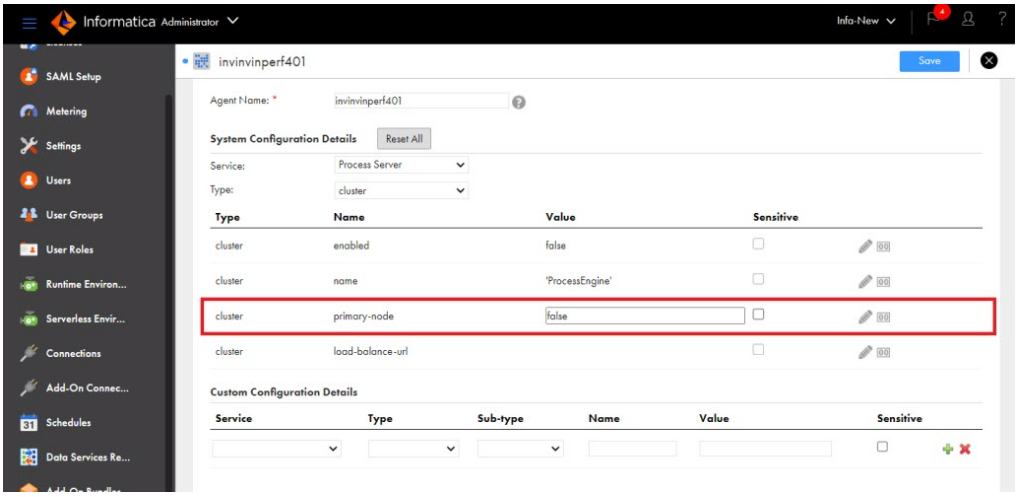
1. In the **Administrator** service, on the **Runtime Environments** page, click the name of the Secure Agent.
2. Click the **Details** tab.
3. In the upper right corner, click **Edit**.
4. To edit a service property, perform the following steps:
   1. In the **System Configuration Details** area, select the service as **Process Server**.

2. Select the configuration property type as **db**.
3. In the row that contains the **Name** as **url**, click the **Edit Agent Configuration** icon and enter the Azure hosted PostgreSQL database url in the `hostname:port_number` format.



Here, `hostname` is the Azure hosted PostgreSQL database server name and `port` is the database port. Ensure that you enter the `localhost` as the `hostname` of the Azure hosted PostgreSQL database server name, and replace `activevos` with your database name, for example, <`user_db`>.

4. Select the configuration property type as **cluster**.
5. In the row that contains the **Name** as **primary-node**, click the **Edit Agent Configuration** icon, and update the value to **false** as shown in the following image:



6. Click **Save**.
   The Process Server will be in the restart required state after making the changes to the system configuration.

5. Add your machine IP address where the Secure Agent is installed to the firewall rules of Networking in the Azure portal.
6. Log in to the Secure Agent machine.
7. Go to the following directory:

`<Secure Agent installation directory>\apps\agentcore`

8. Run `infaagent shutdown`.
9. Go to the following directory:

   `<Secure Agent installation directory>\apps\process-engine\data\db\util`
10. Run `server_stop`.
11. To start the Secure Agent, go to the following directory:

    `<Secure Agent installation directory>\apps\agentcore`
12. Run `infaagent startup`.

    To verify that your user name and password are being used, after the Process Server is up, verify that the `-Dpe.db.username` value is set to your user name in the Catalina logs available in the following directory:

    `<Secure Agent installation directory>\apps\process-engine\logs`

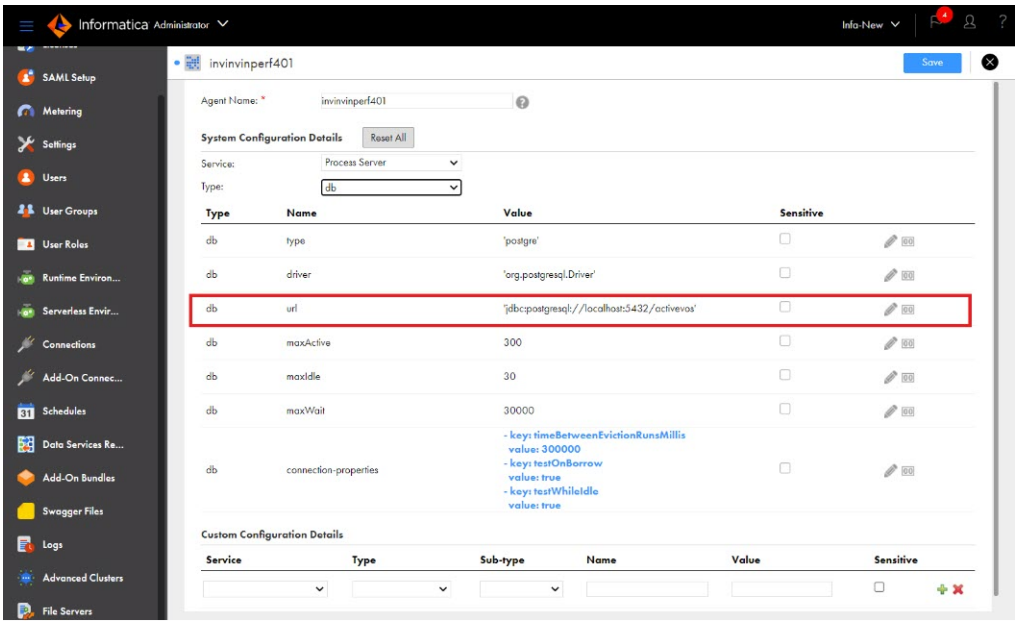
    It might take some time for the Process Server to restart depending upon the latency between the Secure Agent machine and the PostgreSQL database.
13. In the **Application Integration Console** service, on the **Process Schedules** page, enable schedules on the Secure Agent.
14. Validate the processes, connections, and service connectors after the Process Server is connected to the Azure hosted PostgreSQL database.

Roll back to the original cluster database

When migrating data from the existing cluster database dump to the Azure hosted PostgreSQL database, if you face any issue, you can roll back to the original cluster.

To roll back to the original cluster, perform the following steps:

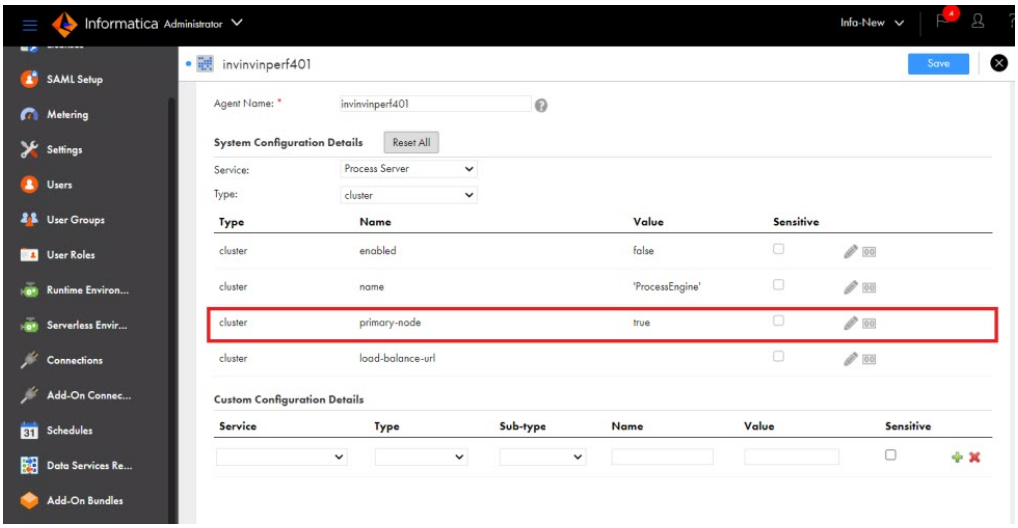
1. In the **Application Integration Console** service, on the **Process Schedules** page, disable the schedules on the Secure Agent.
2. Select all the enabled schedules, and from the drop-down list, click **Disable**.
3. In the **Administrator** service, on the **Runtime Environments** page, click the name of the Secure Agent.
4. Click the **Details** tab.
5. In the **Agent Service Start or Stop** area, select the service as **Process Server** and click **Stop**.
6. In the upper right corner, click **Edit**.
7. To edit a service property, perform the following steps:
   a. In the **System Configuration Details** area, select the service as **Process Server**.
   b. Select the configuration property type as **db**.
   c. In the row that contains the **Name** as **url**, click the **Edit Agent Configuration** icon and enter `hostname:port_number` as shown in the following image:



Here, `hostname` and `port` are the database host and database port. Ensure that you replace the `hostname` of the Azure hosted PostgreSQL database server name with `localhost`, and replace your database name, for example, <`user_db`> with `activevos`.

d. Select the configuration property type as **cluster**.

e. In the row that contains the **Name** as **primary-node**, click the **Edit Agent Configuration** icon, and update the value to **true** as shown in the following image:



Note: If the **primary-node** is set to **false**, the PostgreSQL database does not start on that node.

f. Click **Save**.

The Process Server will be in the restart required state after making the changes to the system configuration.

8. Delete the environment variables `PE_DB_USERNAME` and `PE_DB_PASSWORD`.
9. Log in to the Secure Agent machine.
10. Go to the following directory:

`<Secure Agent installation directory>\apps\agentcore`

11. Run `infaagent shutdown`.
12. Go to the following directory:

    `<Secure Agent installation directory>\apps\process-engine\data\db\util`
13. Run `server_stop`.
14. To start the Secure Agent, go to the following directory:

    `<Secure Agent installation directory>\apps\agentcore`
15. Run `infaagent startup`.
16. In the **Application Integration Console** service, on the **Process Schedules** page, enable the schedules on the Secure Agent.
17. Validate the processes, connections, and service connectors after the database rollback.

Author

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